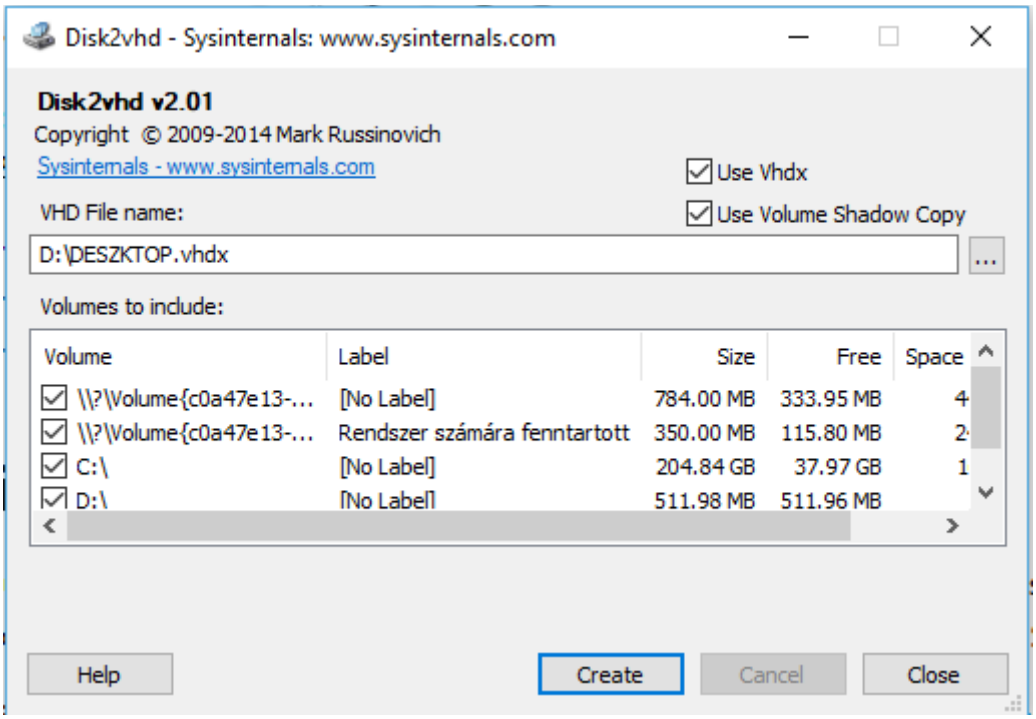
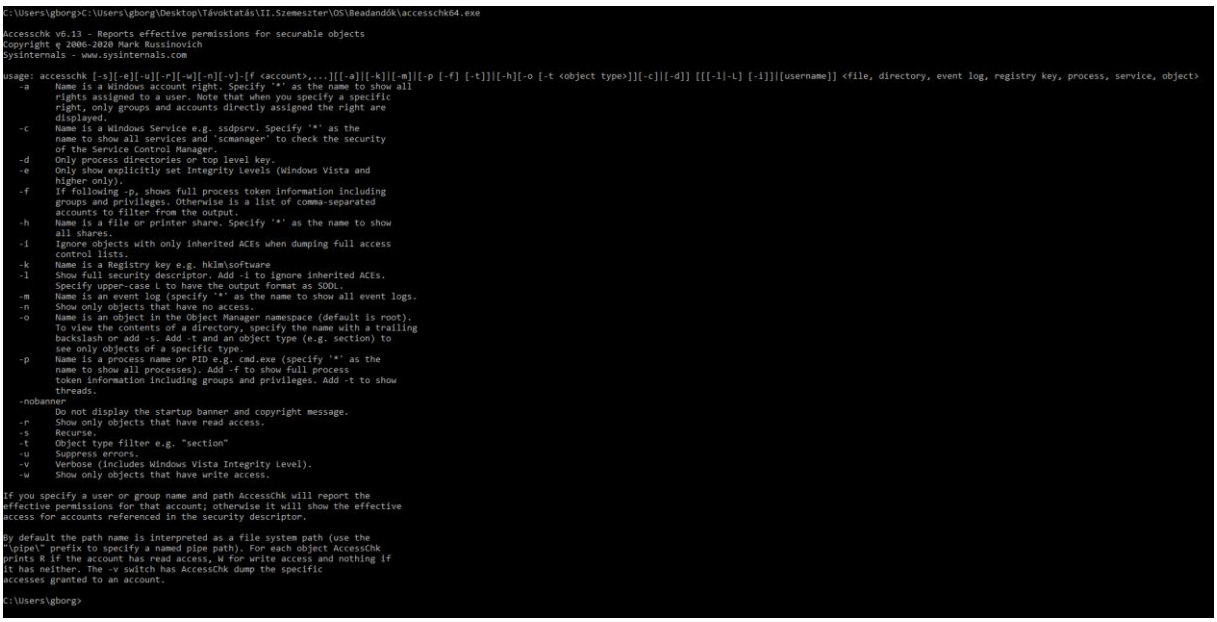
Disk2vhd



- A programmal létrehozhatunk egy virtuális hard disket a meglévő diskjeinkből, amiket fel lehet használni pl a virtuális számítógépekhez.
- A futtatása után kiválaszthatjuk, mely lemezből szeretnénk virtuálisat létrehozni.

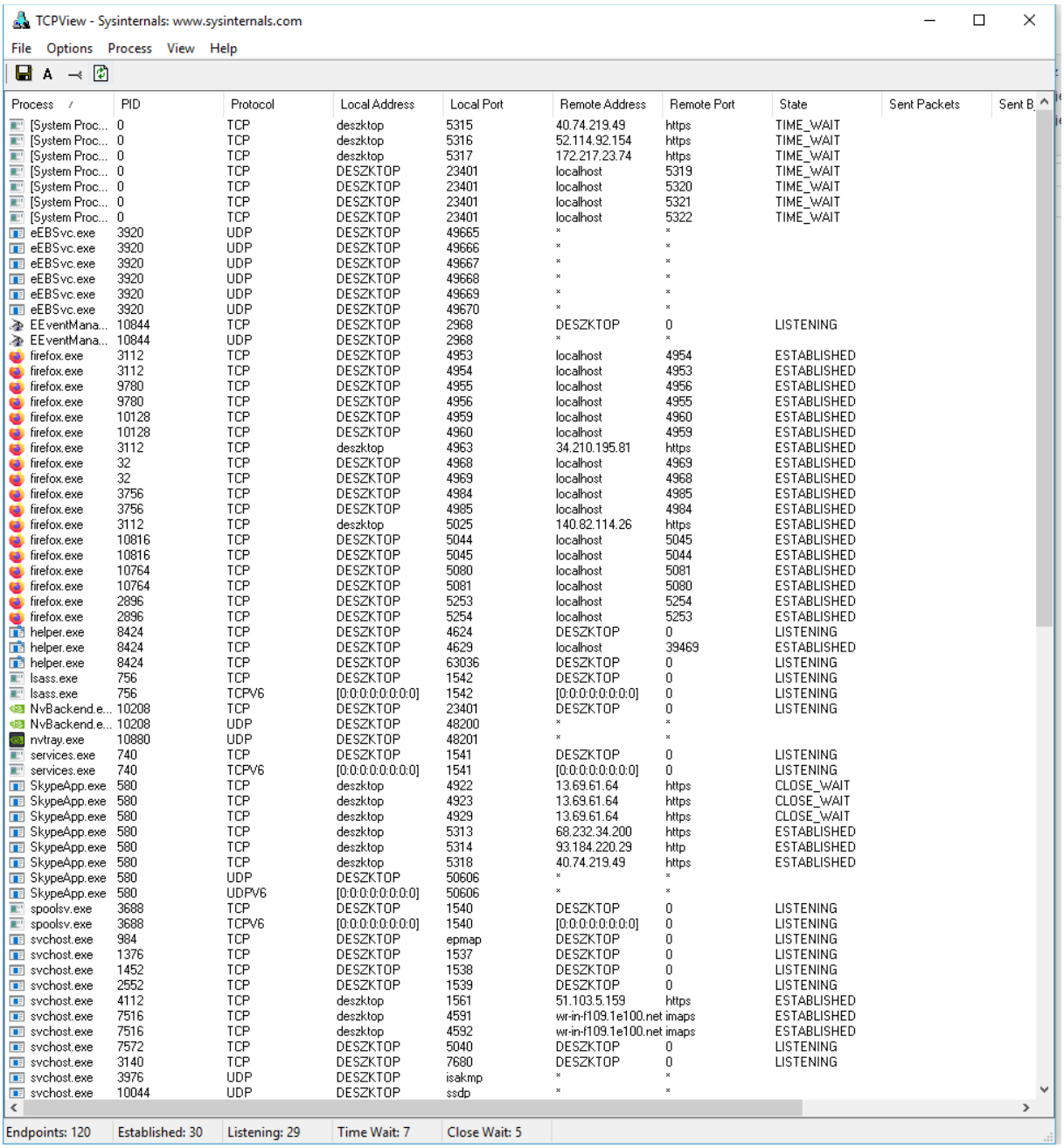
AccessChk



```
C:\Users\gborg>C:\Users\gborg\Desktop\Távoktatás\II.Szemeszter\OS\Beadandók\accesschk64.exe

Accesschk v6.13 - Reports effective permissions for securable objects
Copyright ę 2006-2020 Mark Russinovich
Sysinternals - www.sysinternals.com

usage: accesschk [-s][-e][-u][-r][-w][-n][-v]-[f <account>,...][[-a][[-k][[-m][[-p [-f] [-t]][[-h][-o [-t <object type>]][-c][[-d]] [[[-l|-L] [-i]][[username]] <file, directory, event log, registry key, process, service, object>
   -a    Name is a Windows account right. Specify '*' as the name to show all
         rights assigned to a user. Note that when you specify a specific
         right, only groups and accounts directly assigned the right are
         displayed.
   -c    Name is a Windows Service e.g. ssdpsrv. Specify '*' as the
         name to show all services and `scmanager` to check the security
         of the Service Control Manager.
   -d    Only process directories or top level key.
   -e    Only show explicitly set Integrity Levels (Windows Vista and
         higher only).
   -f    If following -p, shows full process token information including
         groups and privileges. Otherwise is a list of comma-separated
         accounts to filter from the output.
   -h    Name is a file or printer share. Specify '*' as the name to show
         all shares.
   -i    Ignore objects with only inherited ACEs when dumping full access
         control lists.
   -k    Name is a Registry key e.g. hklm\software
   -l    Show full security descriptor. Add -i to ignore inherited ACEs.
         Specify upper-case L to have the output format as SDDL.
   -m    Name is an event log (specify '*' as the name to show all event logs.
   -n    Show only objects that have no access.
   -o    Name is an object in the Object Manager namespace (default is root).
         To view the contents of a directory, specify the name with a trailing
         backslash or add -s. Add -t and an object type (e.g. section) to
         see only objects of a specific type.
   -p    Name is a process name or PID e.g. cmd.exe (specify '*' as the
         name to show all processes). Add -f to show full process
         token information including groups and privileges. Add -t to show
         threads.
   -nobanner
         Do not display the startup banner and copyright message.
   -r    Show only objects that have read access.
   -s    Recurse.
   -t    Object type filter e.g. "section"
   -u    Suppress errors.
   -v    Verbose (includes Windows Vista Integrity Level).
   -w    Show only objects that have write access.

If you specify a user or group name and path AccessChk will report the
effective permissions for that account; otherwise it will show the effective
access for accounts referenced in the security descriptor.

By default the path name is interpreted as a file system path (use the
"\pipe\" prefix to specify a named pipe path). For each object AccessChk
prints R if the account has read access, W for write access and nothing if
it has neither. The -v switch has AccessChk dump the specific
accesses granted to an account.

C:\Users\gborg>
```

- A programmal megtekinthetjük, milyen fájlok érhetőek el a rendszer személyei, csoportjai számára.
- A terminálban való futtatása után paraméterek segítségével használhatunk utasításokat.

TCPView



TCPView - Sysinternals: www.sysinternals.com

File Options Process View Help

Process	PID	Protocol	Local Address	Local Port	Remote Address	Remote Port	State	Sent Packets	Sent B
[System Proc...	0	TCP	deszktop	5315	40.74.219.49	https	TIME_WAIT		
[System Proc...	0	TCP	deszktop	5316	52.114.92.154	https	TIME_WAIT		
[System Proc...	0	TCP	deszktop	5317	172.217.23.74	https	TIME_WAIT		
[System Proc...	0	TCP	DESZKTOP	23401	localhost	5319	TIME_WAIT		
[System Proc...	0	TCP	DESZKTOP	23401	localhost	5320	TIME_WAIT		
[System Proc...	0	TCP	DESZKTOP	23401	localhost	5321	TIME_WAIT		
[System Proc...	0	TCP	DESZKTOP	23401	localhost	5322	TIME_WAIT		
eEBSvc.exe	3920	UDP	DESZKTOP	49665	*	*			
eEBSvc.exe	3920	UDP	DESZKTOP	49666	*	*			
eEBSvc.exe	3920	UDP	DESZKTOP	49667	*	*			
eEBSvc.exe	3920	UDP	DESZKTOP	49668	*	*			
eEBSvc.exe	3920	UDP	DESZKTOP	49669	*	*			
eEBSvc.exe	3920	UDP	DESZKTOP	49670	*	*			
EEventMana...	10844	TCP	DESZKTOP	2968	DESZKTOP	0	LISTENING		
EEventMana...	10844	UDP	DESZKTOP	2968	*	*			
firefox.exe	3112	TCP	DESZKTOP	4953	localhost	4954	ESTABLISHED		
firefox.exe	3112	TCP	DESZKTOP	4954	localhost	4953	ESTABLISHED		
firefox.exe	9780	TCP	DESZKTOP	4955	localhost	4956	ESTABLISHED		
firefox.exe	9780	TCP	DESZKTOP	4956	localhost	4955	ESTABLISHED		
firefox.exe	10128	TCP	DESZKTOP	4959	localhost	4960	ESTABLISHED		
firefox.exe	10128	TCP	DESZKTOP	4960	localhost	4959	ESTABLISHED		
firefox.exe	3112	TCP	deszktop	4963	34.210.195.81	https	ESTABLISHED		
firefox.exe	32	TCP	DESZKTOP	4968	localhost	4969	ESTABLISHED		
firefox.exe	32	TCP	DESZKTOP	4969	localhost	4968	ESTABLISHED		
firefox.exe	3756	TCP	DESZKTOP	4984	localhost	4985	ESTABLISHED		
firefox.exe	3756	TCP	DESZKTOP	4985	localhost	4984	ESTABLISHED		
firefox.exe	3112	TCP	deszktop	5025	140.82.114.26	https	ESTABLISHED		
firefox.exe	10816	TCP	DESZKTOP	5044	localhost	5045	ESTABLISHED		
firefox.exe	10816	TCP	DESZKTOP	5045	localhost	5044	ESTABLISHED		
firefox.exe	10764	TCP	DESZKTOP	5080	localhost	5081	ESTABLISHED		
firefox.exe	10764	TCP	DESZKTOP	5081	localhost	5080	ESTABLISHED		
firefox.exe	2896	TCP	DESZKTOP	5253	localhost	5254	ESTABLISHED		
firefox.exe	2896	TCP	DESZKTOP	5254	localhost	5253	ESTABLISHED		
helper.exe	8424	TCP	DESZKTOP	4624	DESZKTOP	0	LISTENING		
helper.exe	8424	TCP	DESZKTOP	4629	localhost	39469	ESTABLISHED		
helper.exe	8424	TCP	DESZKTOP	63036	DESZKTOP	0	LISTENING		
lsass.exe	756	TCP	DESZKTOP	1542	DESZKTOP	0	LISTENING		
lsass.exe	756	TCPV6	[0:0:0:0:0:0:0:0]	1542	[0:0:0:0:0:0:0:0]	0	LISTENING		
NvBackend.e...	10208	TCP	DESZKTOP	23401	DESZKTOP	0	LISTENING		
NvBackend.e...	10208	UDP	DESZKTOP	48200	*	*			
nvtray.exe	10880	UDP	DESZKTOP	48201	*	*			
services.exe	740	TCP	DESZKTOP	1541	DESZKTOP	0	LISTENING		
services.exe	740	TCPV6	[0:0:0:0:0:0:0:0]	1541	[0:0:0:0:0:0:0:0]	0	LISTENING		
SkypeApp.exe	580	TCP	deszktop	4922	13.69.61.64	https	CLOSE_WAIT		
SkypeApp.exe	580	TCP	deszktop	4923	13.69.61.64	https	CLOSE_WAIT		
SkypeApp.exe	580	TCP	deszktop	4929	13.69.61.64	https	CLOSE_WAIT		
SkypeApp.exe	580	TCP	deszktop	5313	68.232.34.200	https	ESTABLISHED		
SkypeApp.exe	580	TCP	deszktop	5314	93.184.220.29	http	ESTABLISHED		
SkypeApp.exe	580	TCP	deszktop	5318	40.74.219.49	https	ESTABLISHED		
SkypeApp.exe	580	UDP	DESZKTOP	50606	*	*			
SkypeApp.exe	580	UDPV6	[0:0:0:0:0:0:0:0]	50606	*	*			
spoolsv.exe	3688	TCP	DESZKTOP	1540	DESZKTOP	0	LISTENING		
spoolsv.exe	3688	TCPV6	[0:0:0:0:0:0:0:0]	1540	[0:0:0:0:0:0:0:0]	0	LISTENING		
svchost.exe	984	TCP	DESZKTOP	epmap	DESZKTOP	0	LISTENING		
svchost.exe	1376	TCP	DESZKTOP	1537	DESZKTOP	0	LISTENING		
svchost.exe	1452	TCP	DESZKTOP	1538	DESZKTOP	0	LISTENING		
svchost.exe	2552	TCP	DESZKTOP	1539	DESZKTOP	0	LISTENING		
svchost.exe	4112	TCP	deszktop	1561	51.103.5.159	https	ESTABLISHED		
svchost.exe	7516	TCP	deszktop	4591	wr-in-f109.1e100.net	imaps	ESTABLISHED		
svchost.exe	7516	TCP	deszktop	4592	wr-in-f109.1e100.net	imaps	ESTABLISHED		
svchost.exe	7572	TCP	DESZKTOP	5040	DESZKTOP	0	LISTENING		
svchost.exe	3140	TCP	DESZKTOP	7680	DESZKTOP	0	LISTENING		
svchost.exe	3976	UDP	DESZKTOP	isakmp	*	*			
svchost.exe	10044	UDP	DESZKTOP	ssdp	*	*			

Endpoints: 120 | Established: 30 | Listening: 29 | Time Wait: 7 | Close Wait: 5

- A program lehetővé teszi számunkra, hogy kilistázzuk az összes TCP és UDP végpontokat a rendszerünkben.
- Futattás után az összes aktív TCP és UDP végpont megjelenik a címükkel egyetemben.

AdRestore

```
Parancssor - C:\Users\gborg\Desktop\Távoktatás\II.Szemeszter\OS\Beadandók\adrestore64.exe

C:\Users\gborg>C:\Users\gborg\Desktop\Távoktatás\II.Szemeszter\OS\Beadandók\adrestore64.exe

AdRestore v1.2
by Mark Russinovich
Sysinternals - www.sysinternals.com


Enumerating domain deleted objects:

Found 0 items matching search criteria.


C:\Users\gborg>C:\Users\gborg\Desktop\Távoktatás\II.Szemeszter\OS\Beadandók\adrestore64.exe

AdRestore v1.2
by Mark Russinovich
Sysinternals - www.sysinternals.com


Enumerating domain deleted objects:
```

- A program segítségével törölt fájlokat tudunk visszaállítani.
- Futás után kilistázza a törölt fájlokat egy domainen belül.

Process Explorer

Process Explorer - Sysinternals: www.sysinternals.com [DESZKTOP\gborg]

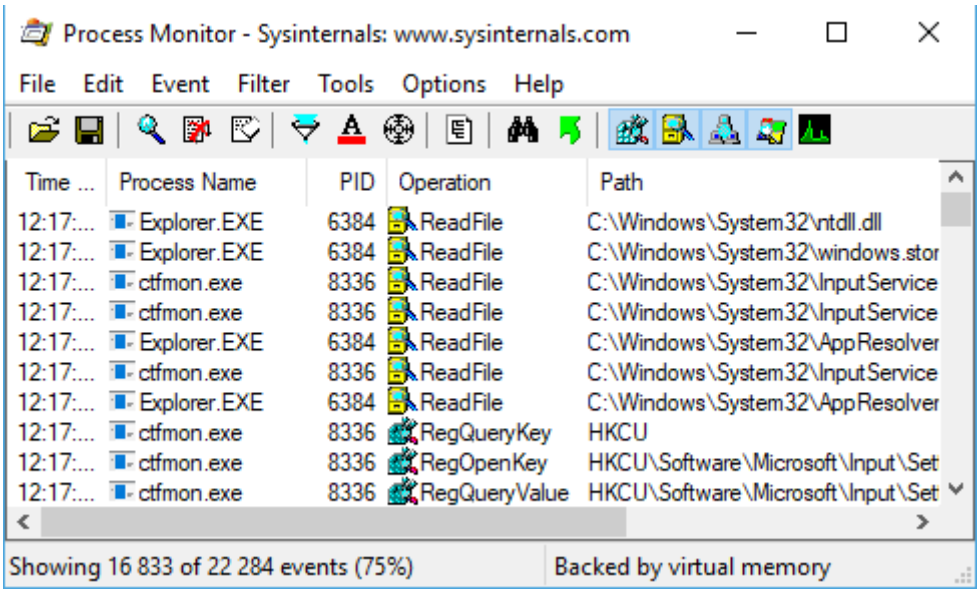
File Options View Process Find Users Help

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Registry		7 988 K	11 536 K	96		
System Idle Process	0.03	52 K	8 K	0		
System	13.92	188 K	144 K	4		
Interrupts	3.56	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		496 K	108 K	336		
Memory Compression	6.01	1 408 K	31 816 K	2372		
csrss.exe	< 0.01	1 780 K	1 380 K	560		
wininit.exe		1 372 K	168 K	664		
services.exe	0.02	5 312 K	4 684 K	740		
svchost.exe		976 K	656 K	872	Windows-szolgáltatások gaz...	Microsoft Corporation
svchost.exe	0.11	12 044 K	14 684 K	908	Windows-szolgáltatások gaz...	Microsoft Corporation
WmiPrvSE.exe		2 680 K	8 532 K	4868		
unsecapp.exe		1 328 K	5 860 K	4984		
dllhost.exe		8 476 K	11 000 K	4584	COM Surrogate	Microsoft Corporation
ShellExperienceHost....	Susp...	29 284 K	536 K	5808	Windows Shell Experience H...	Microsoft Corporation
dllhost.exe		10 468 K	5 556 K	3052	COM Surrogate	Microsoft Corporation
SearchUI.exe	4.64	127 788 K	126 540 K	7400	Search and Cortana applicati...	Microsoft Corporation
RuntimeBroker.exe	< 0.01	8 340 K	12 096 K	7656	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		5 936 K	784 K	7812	Runtime Broker	Microsoft Corporation
SettingSyncHost.exe		8 912 K	3 872 K	8456	Host Process for Setting Syn...	Microsoft Corporation
SkypeBackgroundHo...	Susp...	1 992 K	2 772 K	8852	Microsoft Skype	Microsoft Corporation
Video.UI.exe	Susp...	24 568 K	332 K	8860		
LockApp.exe	Susp...	20 568 K	284 K	9068	LockApp.exe	Microsoft Corporation
RuntimeBroker.exe		8 084 K	2 588 K	4888	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		9 204 K	18 980 K	10352	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		1 976 K	608 K	9736	Runtime Broker	Microsoft Corporation
ApplicationFrameHost...		10 808 K	16 504 K	4248	Application Frame Host	Microsoft Corporation
Microsoft.Photos.exe	Susp...	46 988 K	3 784 K	8716		
RuntimeBroker.exe		12 300 K	4 180 K	7892	Runtime Broker	Microsoft Corporation

CPU Usage: 99.97% | Commit Charge: 77.16% | Processes: 184 | Physical Usage: 89.76%

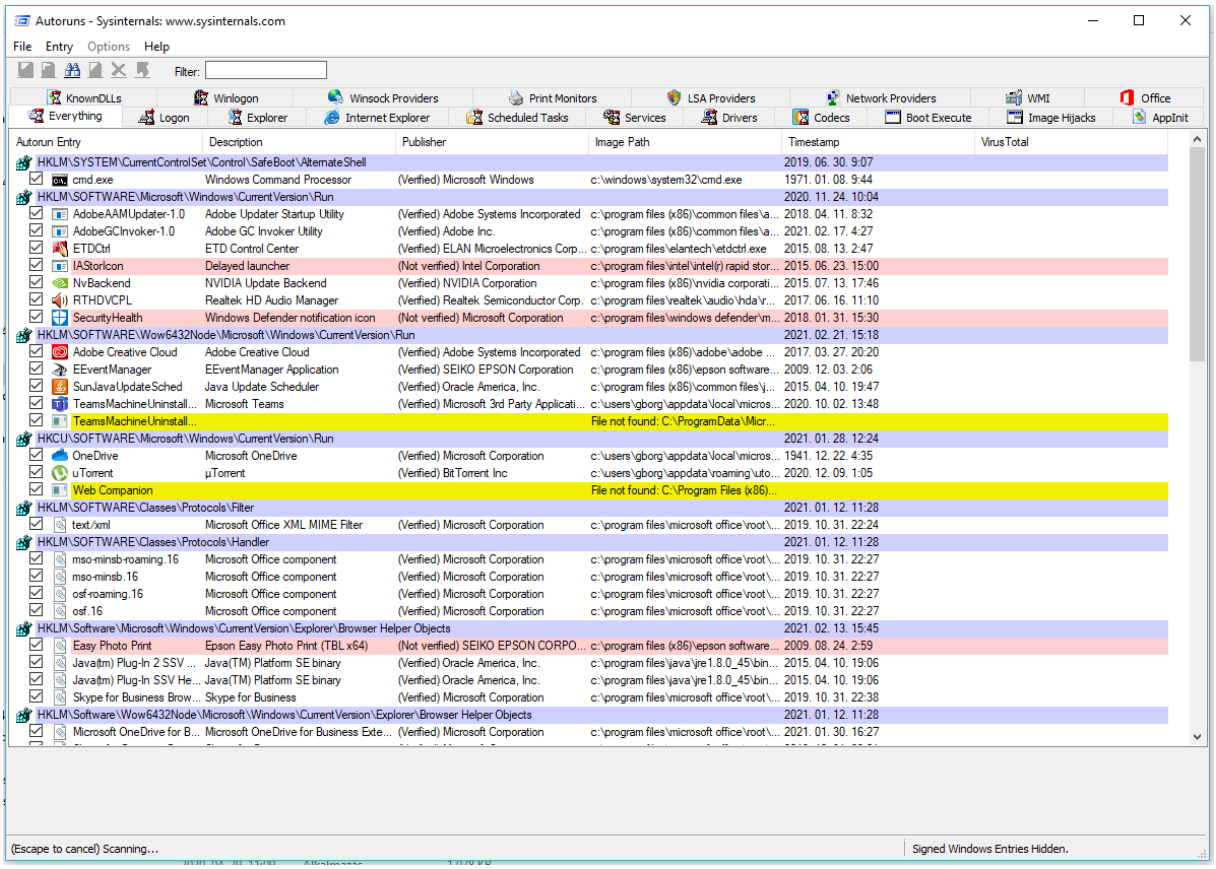
- Segitségével megtudhatjuk, mely programok mely fájlokat vagy könyvtárakat tartják nyitva.
- Futtatás után láthatjuk a programok által futtatott fájlokat és ezeknek méretét/hatását a processzorra.

Process Monitor



- A program valós időben jeleníti meg a fileok, regiszterek és processzek aktvitását.
- Futtatás után láthatjuk a processzek PID számát, rajtuk végzett operációt és az elérési útvonalukat.

AutoRuns



- A program segítségével megtudhatjuk, mely programok futnak le bootoláskor vagy bejelentkezéskor, és mikor indítunk beépített Windows alkalmazásokat.
- Futása után kiírja a programok nevét, futtatás idejét és elérési útvonalát.

Handle

```
Parancssor
  64C: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\ole32.dll.
mui
  650: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\avrt.dll.m
ui
  654: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\setupapi.d
ll.mui
  658: File          C:\Windows\System32\en-US\iphlpapi.dll.mui
  65C: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\pnrpnsp.dl
l.mui
  660: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\dnsapi.dll
.mui
  664: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\dwmapi.dll
.mui
  668: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\ntmarta.dl
l.mui
  66C: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\uxtheme.dl
l.mui
  670: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\dhcpcsvc.d
ll.mui
  674: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\userenv.dl
l.mui
  678: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\napinsp.dl
l.mui
  680: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\webauthn.d
ll.mui
  684: File          C:\Windows\System32\en-US\winrnr.dll.mui
  688: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\DWrite.dll
.mui
----------------------------------------------------------------------------
Teams.exe pid: 4080 DESZKTOP\gborg
   80: File          C:\Users\gborg\AppData\Local\Microsoft\Teams\current
   90: File          C:\Windows\WinSxS\amd64_microsoft.windows.common-controls_6595b64144ccf1df_6.0.17134.829_none_fb46a5473061b9d5
  2DC: File          C:\Users\gborg\AppData\Local\Microsoft\Teams\current\v8_context_snapshot.bin
  2E0: File          C:\Program Files\WindowsApps\Microsoft.LanguageExperiencePackhu-hu_17134.24.32.0_neutral__8wekyb3d8bbwe\Windows\System32\hu-HU\mswsock.dl
l.mui
  2E8: File          C:\Users\gborg\AppData\Local\Microsoft\Teams\current\icudtl.dat
  2F8: File          C:\Users\gborg\AppData\Local\Microsoft\Teams\current\chrome_100_percent.pak
  328: File          C:\Users\gborg\AppData\Local\Microsoft\Teams\current\chrome_200_percent.pak
  330: File          C:\Users\gborg\AppData\Local\Microsoft\Teams\current\locales\hu.pak
  338: File          C:\Users\gborg\AppData\Local\Microsoft\Teams\current\resources.pak
----------------------------------------------------------------------------
SearchProtocolHost.exe pid: 6168 \<unable to open process>
----------------------------------------------------------------------------
SearchFilterHost.exe pid: 11380 \<unable to open process>
----------------------------------------------------------------------------
handle64.exe pid: 10068 DESZKTOP\gborg
   4C: File          C:\Users\gborg
   84: File          C:\Windows\WinSxS\amd64_microsoft.windows.common-controls_6595b64144ccf1df_5.82.17134.829_none_f94fe9573097fa34

C:\Users\gborg>
```

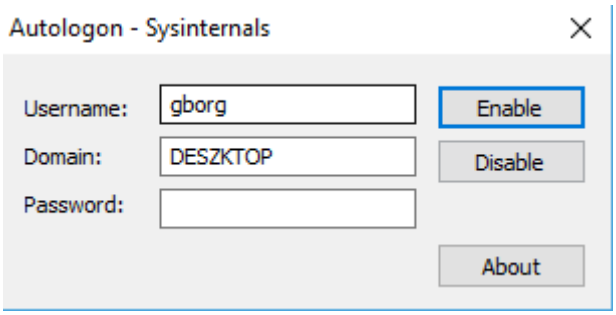
- A parancssorban teszi lehetővé annak megtekintését, mely programok mely fájlokat/könyvtárakat nyitották meg.
- Futás után a parancssorban láthatjuk a megnyitott fájlok elérési útvonalát.

LogonSessions

```
Administrator: Parancssor
    Sid:            S-1-5-96-0-1
    Logon time:     2021. 03. 16. 10:52:19
    Logon server:
    DNS Domain:
    UPN:

[5] Logon session 00000000:00016f36:
    User name:      Window Manager\DWM-1
    Auth package:   Negotiate
    Logon type:     Interactive
    Session:        1
    Sid:            S-1-5-90-0-1
    Logon time:     2021. 03. 16. 10:52:19
    Logon server:
    DNS Domain:
    UPN:

[6] Logon session 00000000:00016f60:
    User name:      Window Manager\DWM-1
    Auth package:   Negotiate
    Logon type:     Interactive
    Session:        1
    Sid:            S-1-5-90-0-1
    Logon time:     2021. 03. 16. 10:52:19
    Logon server:
    DNS Domain:
    UPN:

[7] Logon session 00000000:000003e5:
    User name:      NT AUTHORITY\HELYI SZOLG┴LTAT┴S
    Auth package:   Negotiate
    Logon type:     Service
    Session:        0
    Sid:            S-1-5-19
    Logon time:     2021. 03. 16. 10:52:19
    Logon server:
    DNS Domain:
    UPN:

[8] Logon session 00000000:00022f59:
    User name:      DESZKTOP\gborg
    Auth package:   CloudAP
    Logon type:     Interactive
    Session:        1
    Sid:            S-1-5-21-3706558226-4134771054-4219173159-1001
    Logon time:     2021. 03. 16. 10:52:21
    Logon server:
    DNS Domain:
    UPN:

[9] Logon session 00000000:0002328a:
    User name:      DESZKTOP\gborg
    Auth package:   CloudAP
    Logon type:     Interactive
    Session:        1
    Sid:            S-1-5-21-3706558226-4134771054-4219173159-1001
    Logon time:     2021. 03. 16. 10:52:21
    Logon server:
    DNS Domain:
    UPN:

C:\WINDOWS\system32>
```

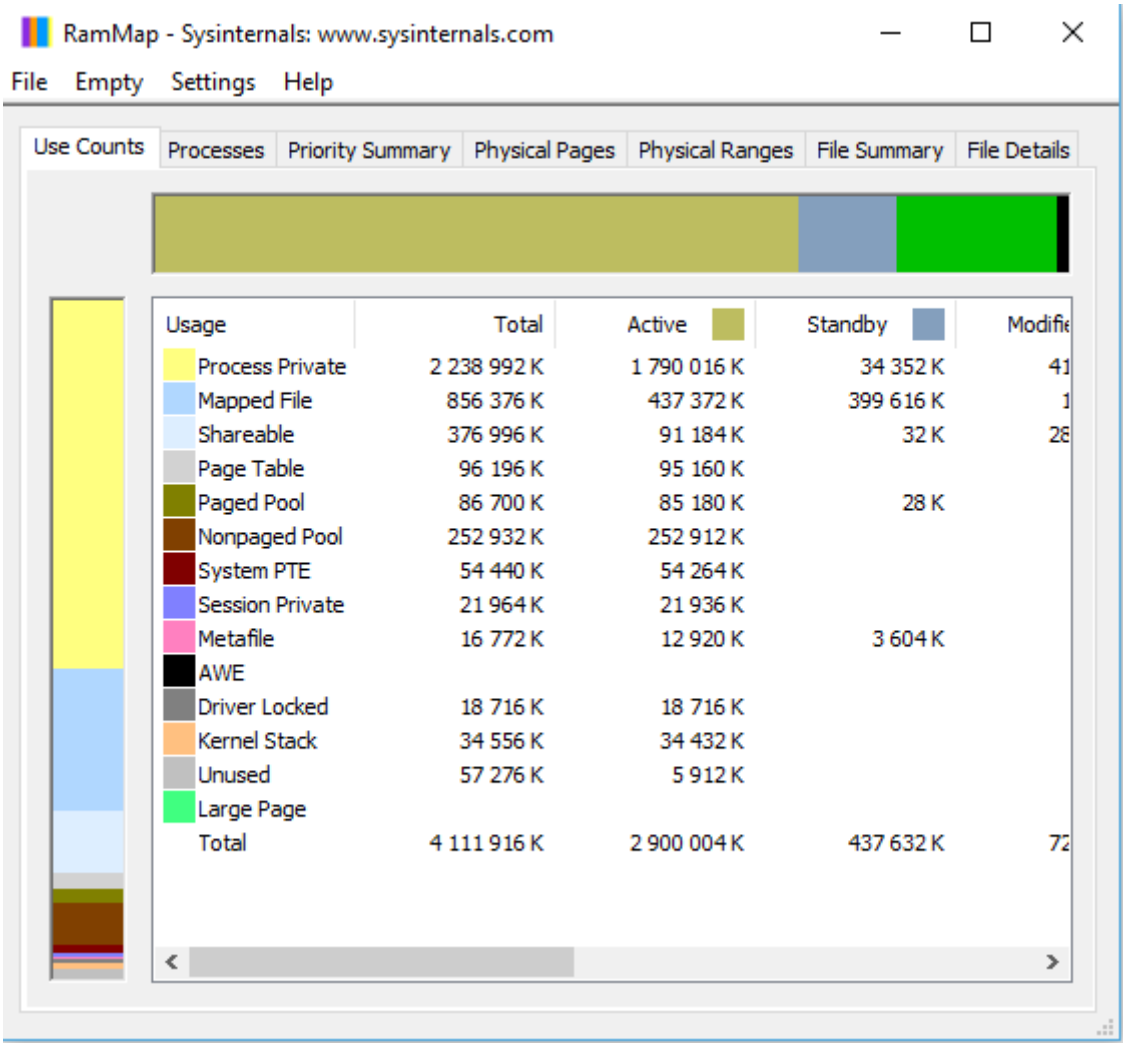
- A bejelentkezéseket kilistázó program.
- Futtatás után kiírja a bejelentkezett munkamenetekhez tartozó címet, nevet, stb…

Autologon



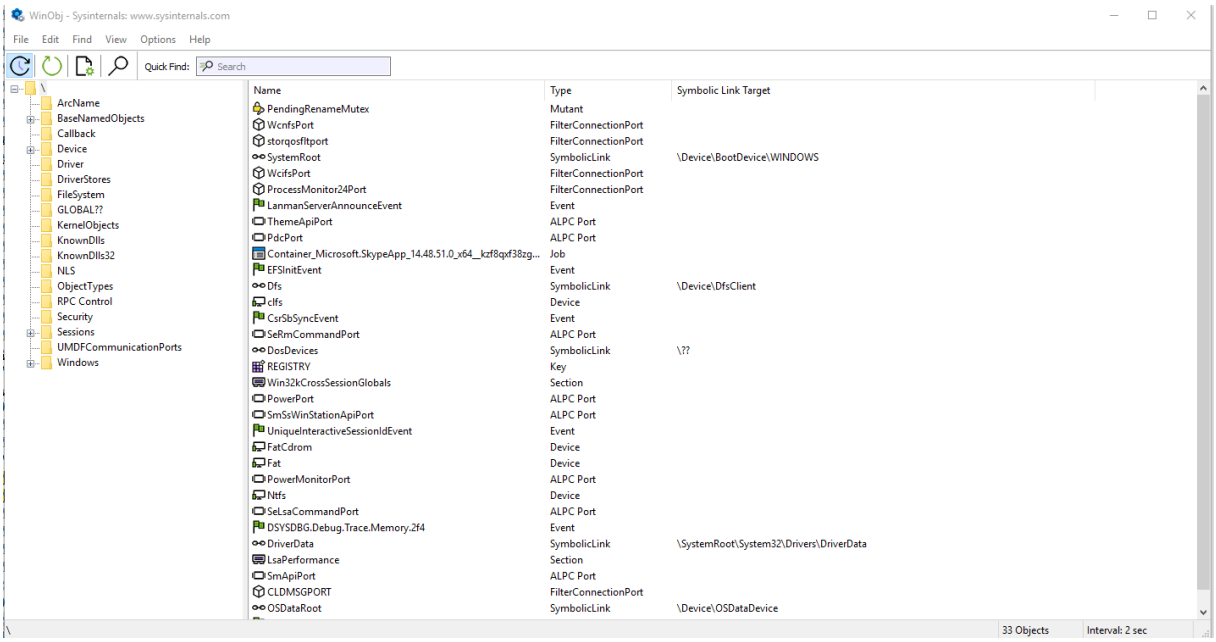
- A rendszerhez tartozó felhasználók adataival lehet bejelentkezni a program segítségével.
- Futtatás után kéri a felhasználó nevét, domainjét és jelszavát, amik megadása után be lehet jelentkezni az adott felhasználóként.

RAMMap



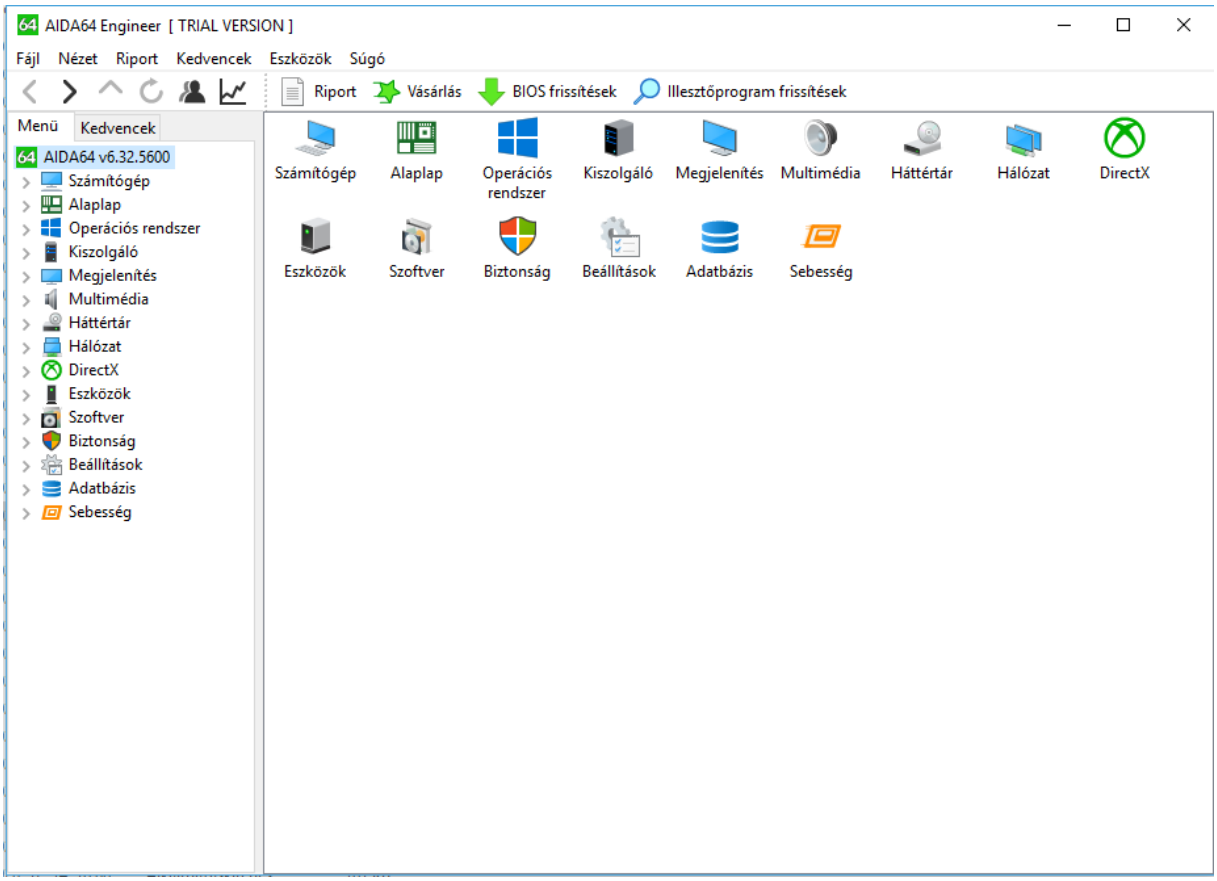
- A program egy memóriahasználat analízisként szolgál.
- Futtatás után láthatjuk a mennyi memória van használatban és ebből mennyi aktív/készenléti.

WinObj



- A program a Windows NT objektek vezérlésére használatos.
- Futtatás után láthatjuk az objekteket, amikkel dolgozhatunk.

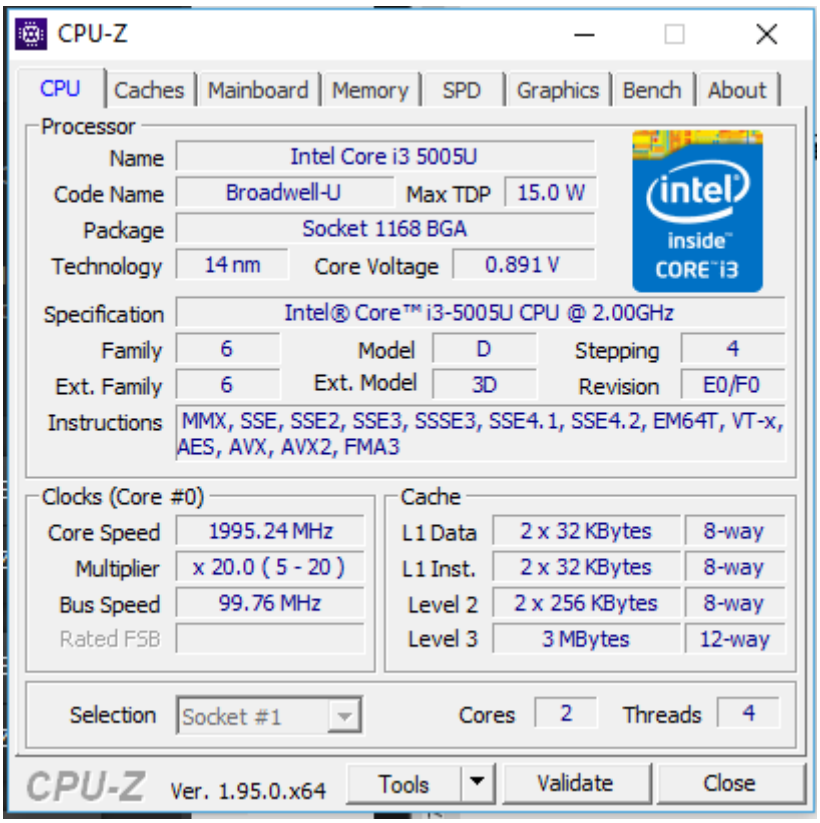
AIDA64



- A program Windows felhasználóknak szánt sebességmérő alkalmazás.

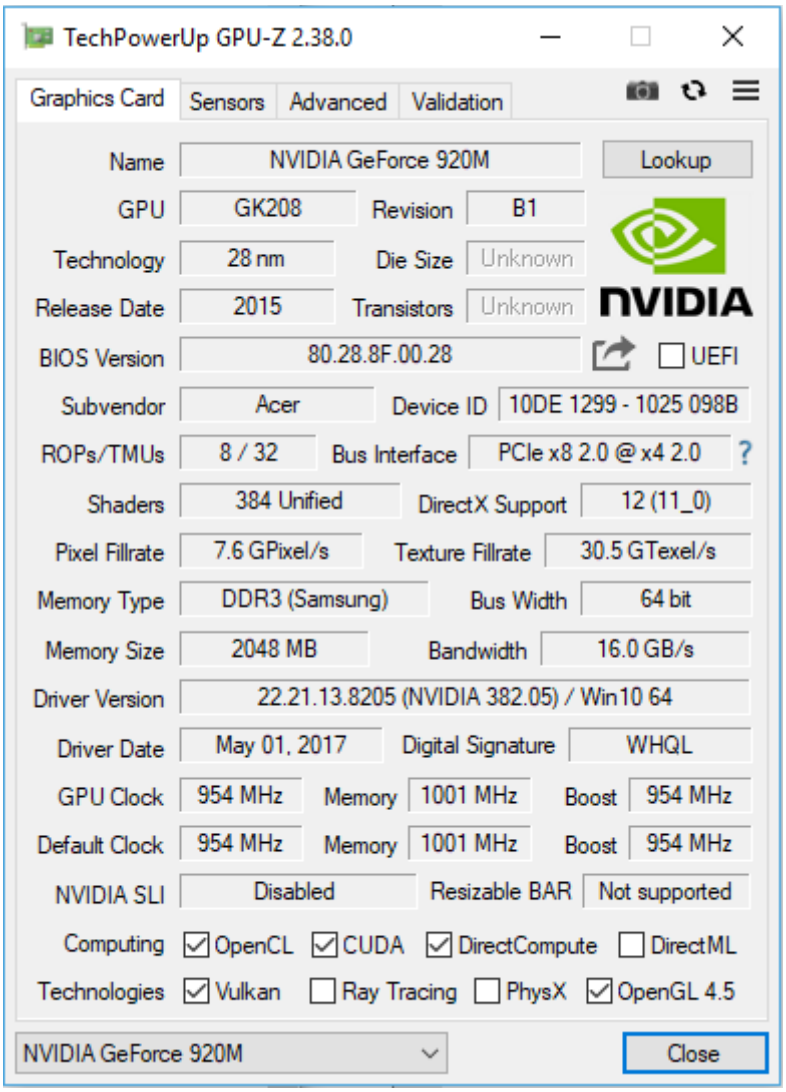
- Futtatás után egy menüsört láthatunk, ahol kiválaszthatjuk, miről szeretnénk diagnosztikát látni.

CPU-Z



- Az alkalmazás a rendszerünk főbb eszközeiről nyújt információt
- Futtatás után láthatjuk ezeket az információkat a CPU-ról, Memóriáról, Grafikus kártyáról, stb…

GPU-Z



- A programmal megtekinthetjük a grafikus kártyánk adatait.
- A futása után minden fontos adat megjelenik a számítógépünk grafikájáról.